

Final Exam, S2 2017 Solutions

QUESTION 1 [20 marks]

(i) [10 marks] For the circuit shown in Figure 1,

a. (8 marks) Apply mesh analysis and show that the mesh equations are given as below,

$$\begin{cases} i_1 + 3i_2 - 2i_3 = 8 \\ i_1 + i_2 - 4i_3 = -1 \\ i_1 - i_2 = -4 \end{cases}$$

b. (2 marks) Given the values of mesh currents as $i_1 = -0.5$ A, $i_2 = 3.5$ A, and $i_3 = 1$ A, find the voltage v across 4-A current source.

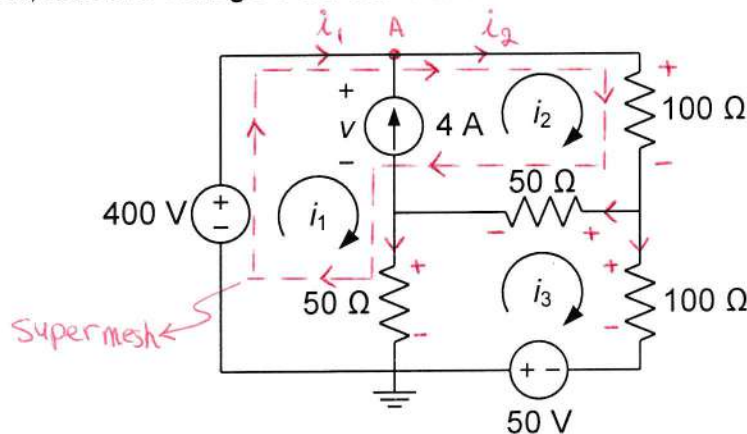


Figure 1

(i)-a:

* KVL in supermesh:

$$-400 + 100i_2 + 50(i_2 - i_3) + 50(i_1 - i_3) = 0 \quad \times \frac{1}{50} \rightarrow$$

$$\rightarrow -8 + 2i_2 + i_2 - i_3 + i_1 - i_3 = 0 \Rightarrow i_1 + 3i_2 - 2i_3 = 8$$

* KVL in mesh 3:

$$-50 - 50(i_1 - i_3) - 50(i_2 - i_3) + 100i_3 = 0 \quad \times \frac{1}{50} \rightarrow$$

$$\rightarrow -1 - i_1 + i_3 - i_2 + i_3 + 2i_3 = 0 \Rightarrow i_1 + i_2 - 4i_3 = -1$$

* KCL at node A:

$$i_1 + 4 = i_2 \Rightarrow i_1 - i_2 = -4$$

(i)-b:

* KVL in mesh 1:

$$v + 50(i_1 - i_3) - 400 = 0 \quad \begin{matrix} i_1 = -0.5 \text{ A} \\ i_3 = 1 \text{ A} \end{matrix} \rightarrow v = 50(i_3 - i_1) + 400$$

$$\Rightarrow v = 50(1 + 0.5) + 400 = 475 \text{ V}$$

Question 1 continued

(ii) [10 marks] For the circuit shown in Figure 2,

a. (8 marks) Apply nodal analysis and show that the nodal equations are given as below,

$$\begin{cases} 11v_1 - 6v_2 = 108 \\ 4v_1 - 3v_2 = -54 \end{cases}$$

b. (2 marks) Given the values of node voltages as $v_1 = 72 \text{ V}$ and $v_2 = 114 \text{ V}$, Calculate the power of the dependent current source.

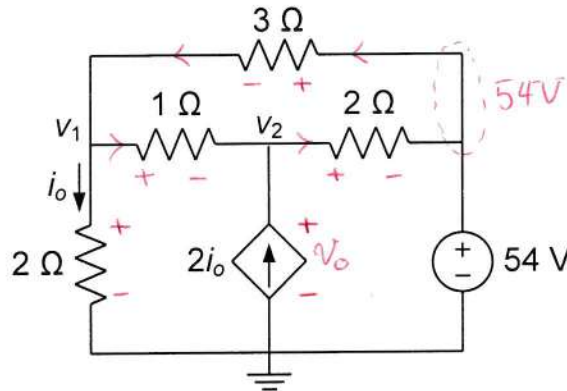


Figure 2

(ii)-a:

* KCL at node 1:

$$\frac{54 - v_1}{3} = \frac{v_1 - 0}{2} + \frac{v_1 - v_2}{1} \quad \times 6 \rightarrow 108 - 2v_1 = 3v_1 + 6v_1 - 6v_2$$

$$\Rightarrow 11v_1 - 6v_2 = 108$$

* KCL at node 2:

$$\frac{v_1 - v_2}{1} + 2i_0 = \frac{v_2 - 54}{2} \quad \left[i_0 = \frac{v_1}{2} \right] \rightarrow v_1 - v_2 + v_1 = \frac{v_2 - 54}{2} \quad \times 2,$$

$$\Rightarrow 4v_1 - 2v_2 = v_2 - 54 \Rightarrow 4v_1 - 3v_2 = -54$$

(ii)-b:

$P_o = 2i_o \times v_o \rightarrow$ passive sign convention for sources (Active element)

$$2i_o = 2 \times \frac{v_1}{2} = v_1 = 72 \text{ A}$$

$$v_o = v_2 = 114 \text{ V}$$

$$\Rightarrow P_o = 72 \times 114 = 8208 \text{ W} = 8.208 \text{ kW Supplied}$$

QUESTION 2 [20 marks]

(i) [10 marks] For the circuit shown in Figure 3,

- (4 marks) Calculate the open-circuit voltage v_{oc} and short-circuit current i_{sc} at the terminals $a-b$.
- (4 marks) Obtain the Norton equivalent circuit with respect to the terminals $a-b$ and draw the equivalent circuit.
- (2 marks) Determine the value of the load resistance R_L for maximum power transfer, and then calculate the maximum power that can be delivered to R_L .

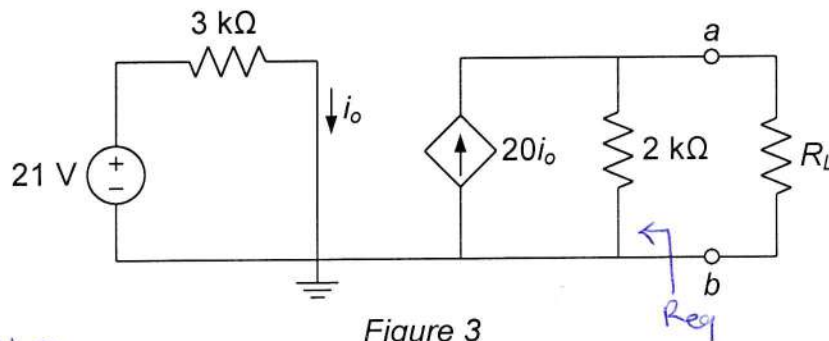
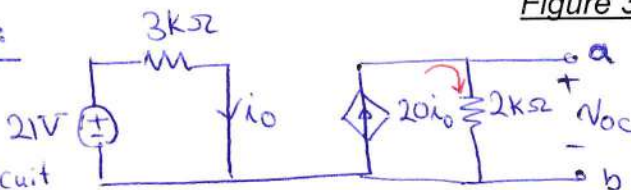


Figure 3

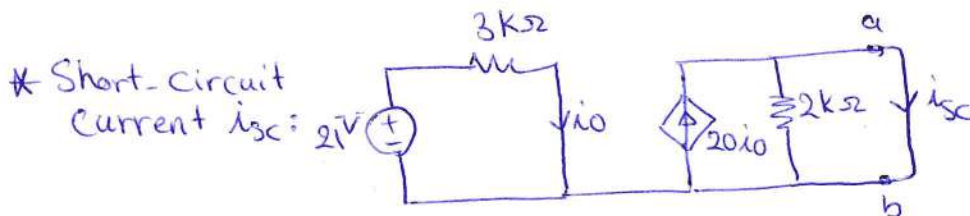
(i)-a:



$$i_o = \frac{21V}{3k} = 7mA \quad \text{①}$$

$$v_{oc} = v_{2k\Omega} = 2k \times 20i_o = 280V$$

* Open-circuit Voltage v_{oc} :

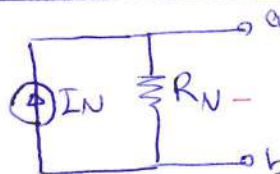


$$i_{sc} = 20i_o \quad \text{①} \quad i_{sc} = 140mA$$

* Short-circuit Current i_{sc} :

(i)-b:

The equivalent circuit Norton



$$\begin{cases} R_N = R_{eq} \\ I_N = i_{sc} \end{cases}$$

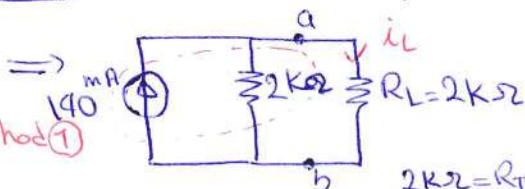
Since there exists a dependent source:

$$R_{eq} = \frac{v_{oc}}{i_{sc}} = \frac{280V}{140mA} = 2k\Omega = R_N$$

$$I_N = i_{sc} = 140mA$$

(i)-c:

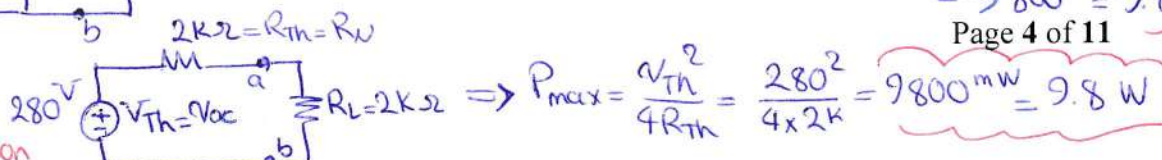
For maximum power transfer $R_L = R_{Th} = R_N = 2k\Omega$



$$i_L = \frac{2k}{2k+2k} \times 140mA = 70mA \Rightarrow P_{max} = R_L i_L^2 = 2k \times (70mA)^2 = 9800mW = 9.8W$$

* Method ①

* Method ②
Source transformation



$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{280^2}{4 \times 2k} = 9800mW = 9.8W$$

Question 2 continued

(ii) [10 marks] In circuit shown in Figure 4, the switch has been in position A for a long time. At $t = 0$, the switch moves to position B.

- (4 marks) Find the voltage $v(t)$ across the capacitor immediately after the switch changes to position B, $v(0^+)$, and its final voltage when $t \rightarrow \infty$, $v(\infty)$.
- (4 marks) Derive an expression for the capacitor voltage $v(t)$ for all time (i.e., for both $t < 0$ and $t > 0$).
- (2 marks) Find the current $i(t)$ through $3\text{-k}\Omega$ resistor for $t > 0$.

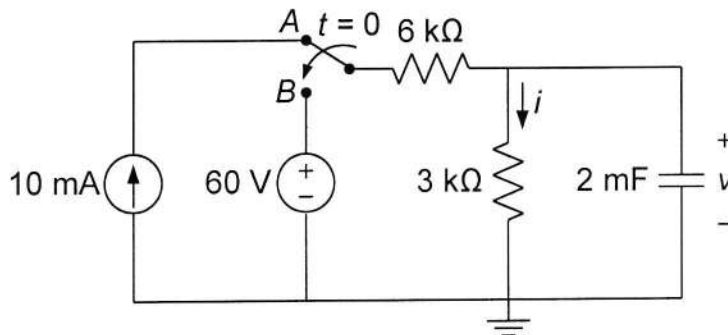
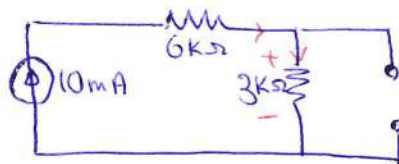


Figure 4

(ii)-a:

$v(0^+) = v(0^-)$ No sudden change in capacitor voltage

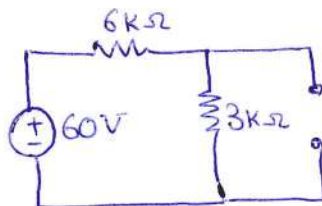
* Circuit when $t < 0$
Switch is at A



$$v(0^-) \Rightarrow v(0^-) = V_{3k\Omega} = 3k \times 10^m = 30V$$

$$\Rightarrow v(0^+) = v(0^-) = 30V$$

* Circuit when $t > 0$
and $t \rightarrow \infty$
Switch is at B

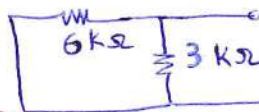


$$v(\infty) = \frac{3k}{6k + 3k} \times 60 = 20V$$

Voltage division

(ii)-b: After switch changes to B $\rightarrow v(t) = v(\infty) + (v(0^+) - v(\infty))e^{-\frac{t}{R_{Th}C}}$

$$\tau = R_{Th}C \rightarrow$$



$$R_{Th} = 3k \parallel 6k = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2k\Omega \Rightarrow \tau = 2k\Omega \times 2mF = 4s$$

$$\Rightarrow v(t) = \begin{cases} 30V & t \leq 0 \\ 20 + 10e^{-\frac{t}{4}} V & t \geq 0 \end{cases}$$

$$\Rightarrow v(t) = 20 + (30 - 20)e^{-\frac{t}{4}} = 20 + 10e^{-\frac{t}{4}} V$$

(ii)-c: $i(t) = \frac{V_{3k\Omega}}{3k\Omega} = \frac{v(t)}{3k\Omega}$

$3k\Omega$ is in parallel with Capacitor

$$t < 0 \rightarrow i(t) = \frac{30}{3k\Omega} = 10mA$$

$$t \geq 0 \rightarrow i(t) = \frac{20}{3} + \frac{10}{3}e^{-\frac{t}{4}} mA$$

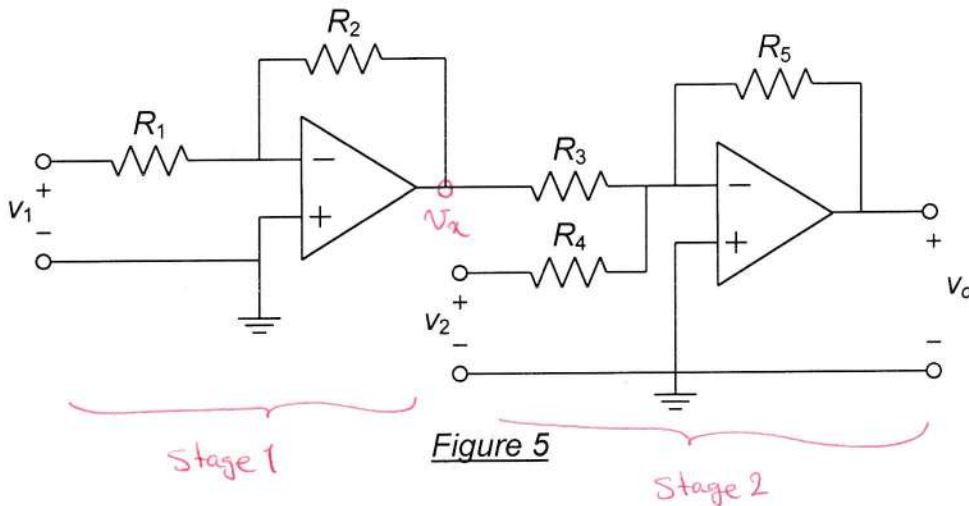
$$\Rightarrow i(t) = \begin{cases} 10mA & t \leq 0 \\ \frac{20}{3} + \frac{10}{3}e^{-\frac{t}{4}} mA & t \geq 0 \end{cases}$$

QUESTION 3 [20 marks]

- (i) [8 marks] For the Op Amp circuit shown in Figure 5 show that the output v_o can be given by the following equation,

$$v_o = a_1 v_1 - a_2 v_2$$

and determine the parameters a_1 and a_2 in terms of resistors R_1 to R_5 .



(i):

* Stage 1: Inverting amplifier:

$$\frac{v_x}{v_1} = - \frac{R_2}{R_1} \quad \text{(I)}$$

* Stage 2: Summing amplifier:

$$v_o = - \left(\frac{R_5}{R_3} v_x + \frac{R_5}{R_4} v_2 \right) \xrightarrow{\text{(I)}} v_o = - \left(\frac{R_5}{R_3} \times \frac{R_2}{R_1} v_1 + \frac{R_5}{R_4} v_2 \right)$$

$$\Rightarrow v_o = \frac{R_5 R_2}{R_3 R_1} v_1 - \frac{R_5}{R_4} v_2$$

$$\Rightarrow v_o = a_1 v_1 - a_2 v_2$$

choose

$$\begin{cases} a_1 = \frac{R_5 R_2}{R_3 R_1} \\ a_2 = \frac{R_5}{R_4} \end{cases}$$

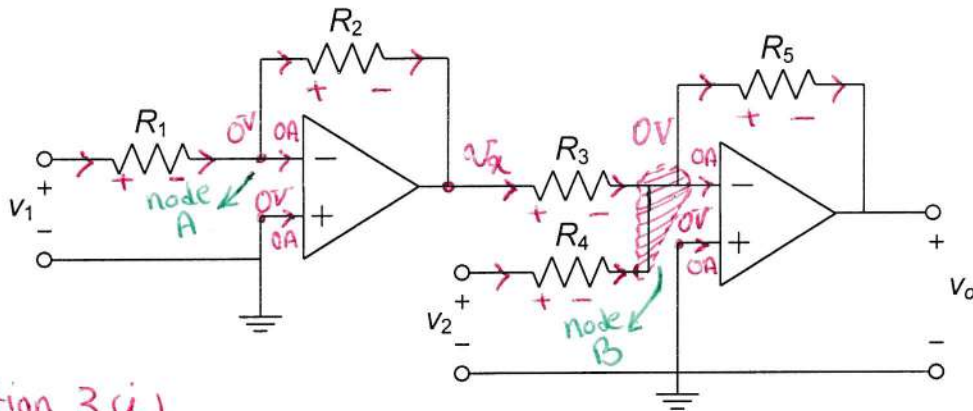


Figure 5

Question 3(i)

Nodal analysis approach:

* KCL at node A: $\frac{V_1 - 0}{R_1} = \frac{0 - v_x}{R_2} \xrightarrow{\times -R_2} v_x = -\frac{R_2}{R_1} V_1$ (I)

* KCL at node B: $\frac{v_x - 0}{R_3} + \frac{V_2 - 0}{R_4} = \frac{0 - V_0}{R_5} \xrightarrow{\times -R_5} V_0 = -\frac{R_5}{R_3} v_x - \frac{R_5}{R_4} V_2$ (II)

(I) $\rightarrow$ (II)
Substitute v_x

$$V_0 = +\frac{R_5 R_2}{R_3 R_1} V_1 - \frac{R_5}{R_4} V_2$$

choose

$$a_1 = \frac{R_5 R_2}{R_3 R_1}$$

$$a_2 = \frac{R_5}{R_4}$$

$$\Rightarrow V_0 = a_1 V_1 - a_2 V_2$$

Question 3 continued

(ii) [12 marks] In the circuit of Figure 6, the switch has been in position A for a long time before changing its position to B at $t = 0$.

- a. (10 marks) Derive an expression for the inductor current $i_L(t)$ for all time (i.e., for both $t < 0$ and $t > 0$) and sketch $i_L(t)$ as a function of time showing all critical points in the sketch.
- b. (2 marks) Find the voltage $v(t)$ across the $2\text{-}\Omega$ resistor for $t > 0$.

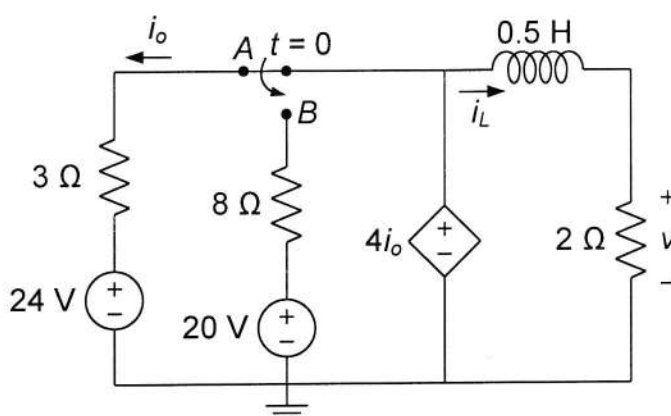
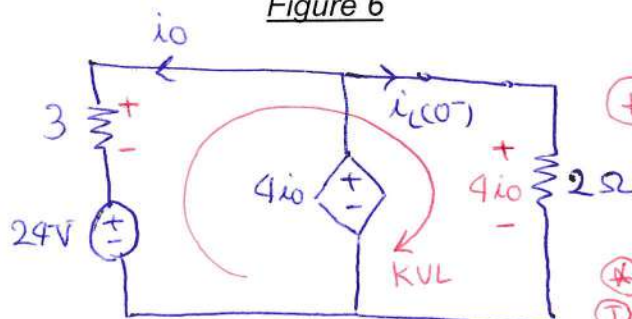


Figure 6

(ii)-a:

Circuit at $t < 0$
Switch is at A



* KVL in big loop to find i_o

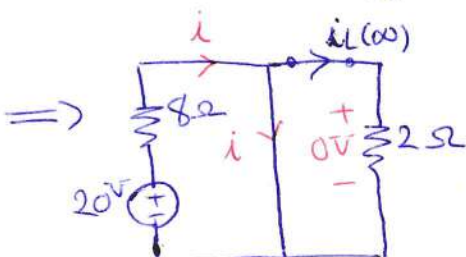
$$-24 - 3i_o + 4i_o = 0 \rightarrow i_o = 24\text{A}$$

Using (II) $i_L(0^-) = 2i_o = 48\text{A}$ $t < 0$

* $2\text{-}\Omega$ resistor is in parallel with dependent voltage source thus $V_{2\Omega} = 4i_o$ (I)

* Also $V_{2\Omega} = 2i_L(0^-)$
(I) $\Rightarrow$ Thus $4i_o = 2i_L(0^-)$
 $\Rightarrow i_L(0^-) = 2i_o$ (II)

Circuit at $t > 0$
and $t \rightarrow \infty$
Switch is at B



$$\Rightarrow i_L(\infty) = \frac{0}{2} = 0\text{A} \quad t \rightarrow \infty$$

When switch changes to B, $i_o = 0\text{A}$ which means any dependent source relying on i_o will be zero.

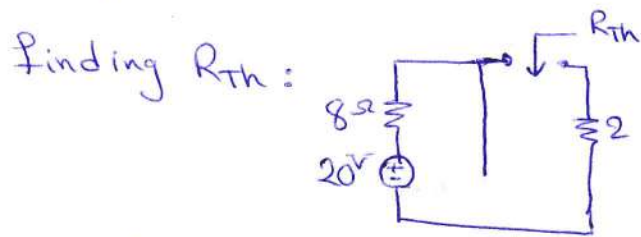
$$i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-\frac{t}{\tau}}, \quad \tau = \frac{L}{R_{th}}$$

$$i_L(0^+) = i_L(0^-) = 48\text{A}$$

No sudden change in inductor current

$\rightarrow$ next page

Q3 part (ii)-a Continued

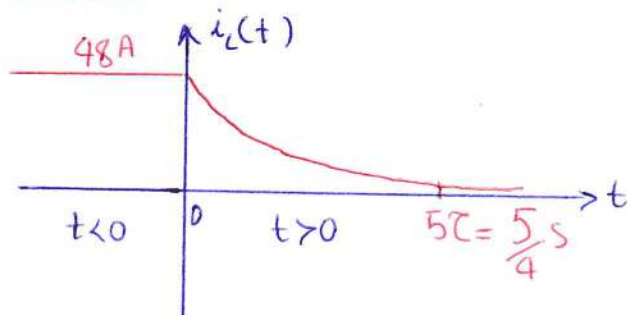


$$R_{Th} = 2\Omega \Rightarrow \tau = \frac{L}{R_{Th}} = \frac{0.5}{2} = \frac{1}{4} s$$

$$\Rightarrow i_L(t) = 0 + [48 - 0] e^{-\frac{t}{\tau}} = 48 e^{-4t} \text{ A } t \geq 0$$

Thus:

$$i_L(t) = \begin{cases} 48 \text{ A} & t \leq 0 \\ 48 e^{-4t} \text{ A} & t \geq 0 \end{cases}$$



(ii)-b: $v(t) = 2i_L(t)$ for all time

$$\Rightarrow v(t) = 2 \times 48 = 96 \text{ V } t \leq 0$$

$$v(t) = \underbrace{2 \times 48}_{96} e^{-4t} \text{ V } t \geq 0$$

$$\Rightarrow v(t) = \begin{cases} 96 \text{ V} & t \leq 0 \\ 96 e^{-4t} \text{ V} & t \geq 0 \end{cases}$$

QUESTION 4 [20 marks]

(i) [10 marks] For the circuit shown in Figure 7,

- (4 marks) Find the equivalent impedance Z_{eq} seen from terminals a-b
- (6 marks) Apply phasor analysis and source transformation to calculate the voltage v_x across the $80\text{-}\Omega$ resistor.

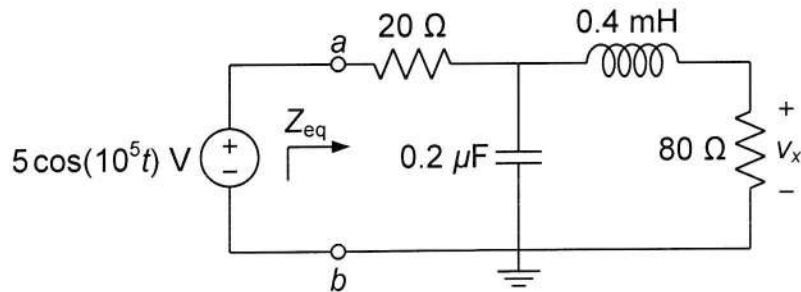


Figure 7

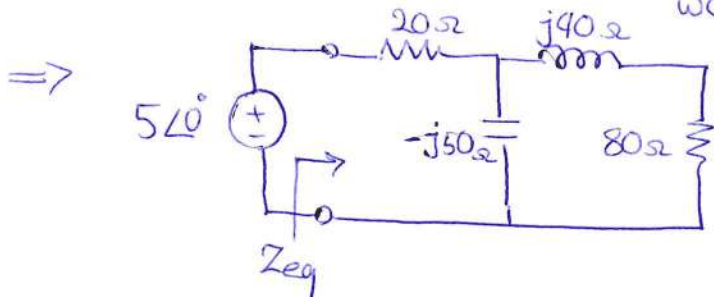
(i)-a:

* Draw the circuit in phasor domain

$$5 \cos(10^5 t) \rightarrow 5 \angle 0^\circ$$

$$\omega = 10^5 \text{ rad/s} \Rightarrow 0.4 \text{ mH} \rightarrow j\omega L = j(10^5 \times 0.4 \times 10^{-3}) = j40 \Omega$$

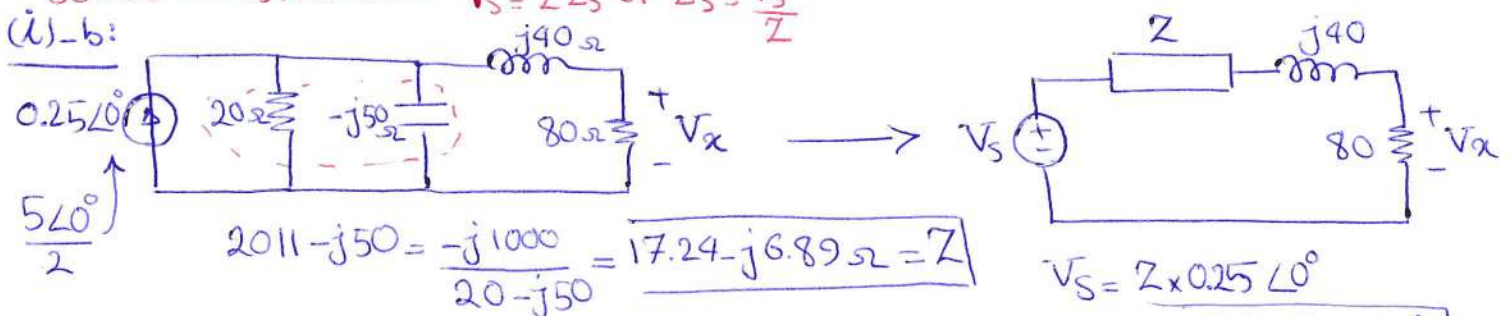
$$0.2 \mu\text{F} \rightarrow \frac{-j}{\omega C} = \frac{-j}{10^5 \times 0.2 \times 10^{-6}} = -j50 \Omega$$



$$Z_{eq} = (80 + j40) \parallel (-j50) + 20$$

$$\Rightarrow Z_{eq} = \frac{(80 + j40) \times (-j50)}{80 + j40 - j50} + 20 = 50.769 - j46.153 \Omega = 68.61 \angle -42.27^\circ \Omega$$

Source transformation $V_s = Z I_s$ or $I_s = \frac{V_s}{Z}$



$$20 \parallel -j50 = \frac{-j1000}{20 - j50} = 17.24 - j6.89 \Omega = Z$$

$$V_s = Z \times 0.25 \angle 0^\circ = 4.642 \angle -21.8^\circ \text{ V}$$

$$\Rightarrow V_x = \frac{80}{Z + j40 + 80} \times V_s = 3.615 \angle -40.6^\circ \text{ V}$$

$$\Rightarrow v_x(t) = 3.615 \cos(10^5 t - 40.6^\circ) \text{ V}$$

Voltage division

Question 4 continued

(ii) [10 marks] For the circuit shown in Figure 8,

- (6 marks) Find the Thevenin equivalent circuit as seen from the terminals $a-b$, and draw the equivalent circuit.
- (2 marks) Determine the value of the load impedance Z_L for maximum average power transfer.
- (2 marks) Calculate the maximum average power that can be delivered to the load Z_L from this circuit.

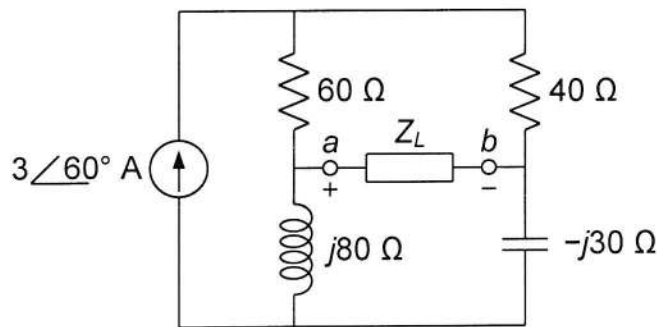
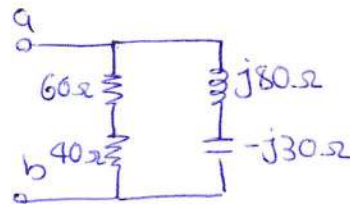
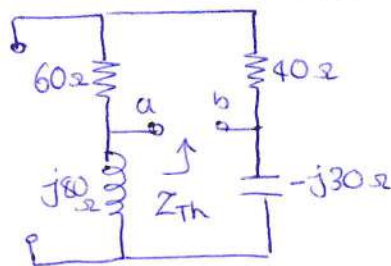


Figure 8

(ii)-a:

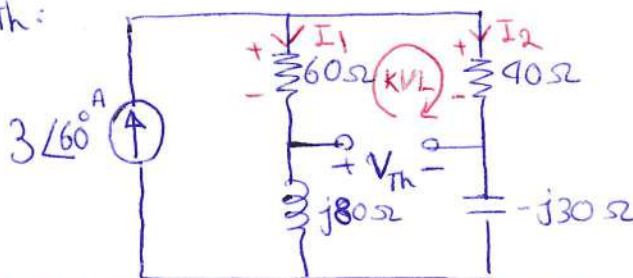
* Find Z_{Th} :



$$\Rightarrow Z_{Th} = (60 + j80) \parallel (40 - j30) = \frac{100 \times j50}{100 + j50} = 20 + j40 \Omega = 44.72 \angle 63.43^\circ \Omega$$

* Find $\bar{V}_{Th}$:

$$\bar{V}_{Th} = V_{oc}$$



Current division

$$\begin{cases} I_1 = \frac{40 - j30}{60 + j80 + 40 - j30} \times 3 \angle 60^\circ \\ I_2 = \frac{60 + j80}{60 - j80 + 40 - j30} \times 3 \angle 60^\circ \end{cases}$$

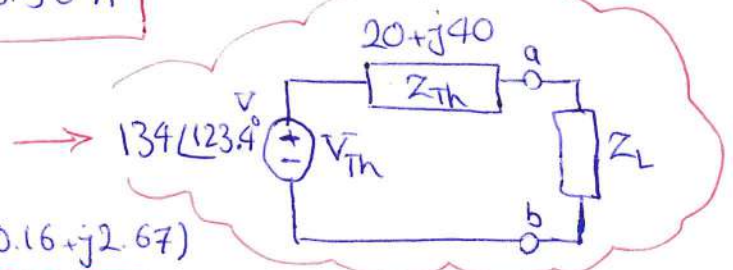
$$\Rightarrow \begin{aligned} I_1 &= 1.34 - j0.08 \text{ A} = 1.34 \angle -3.43^\circ \text{ A} \\ I_2 &= 0.16 + j2.67 \text{ A} = 2.68 \angle 86.56^\circ \text{ A} \end{aligned}$$

* KVL in upper-right mesh:

$$-V_{Th} = 60I_1 + 40I_2 = 0$$

$$\Rightarrow V_{Th} = -60 \times (1.34 - j0.08) + 40 \times (0.16 + j2.67)$$

$$V_{Th} = -74 + j111.6 \text{ V} = 134 \angle 123.4^\circ \text{ V}$$



Q4 part (ii) Continued

(ii)-b: For maximum average power transfer $Z_L = Z_{Th}^*$

Thus $Z_L = 20 - j40 \Omega = 44.72 \angle -63.43^\circ \Omega$

(ii)-c:

$$P_{max} = \frac{|V_{Th}|^2}{8 R_{Th}} = \frac{(134)^2}{8 \times 20} = 112.225 \text{ W}$$

$$Z_{Th} = 20 + j40 \Omega$$

$$\Rightarrow R_{Th} = 20 \Omega$$

QUESTION 5 [20 marks]

(i) [10 marks] For the circuit shown in Figure 9,

- [6 marks] Find the output voltage v_x using phasor analysis and superposition principle.
- [4 marks] Sketch the phasors of voltage source V and current source I along with phasor of voltage V_x on a phasor diagram.

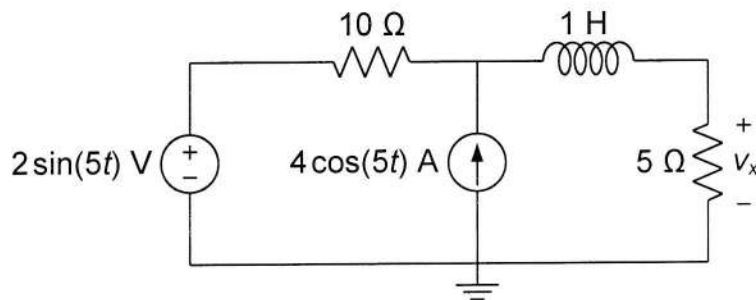


Figure 9

(i) - a:

Both sources have the same frequency

$$V = 2 \sin(5t) = 2 \cos(5t - 90^\circ) = 2 \angle -90^\circ \text{ V} = -2j$$

$$I = 4 \cos(5t) = 4 \angle 0^\circ \text{ A}$$

$$\omega = 5 \rightarrow 1 \text{ H} \rightarrow j\omega L = j5 \Omega$$

$$V_x = V_{x1} + V_{x2} \quad \begin{cases} V_{x1} \text{ is due to voltage source} \\ V_{x2} \text{ is due to current source} \end{cases}$$

same frequency $\rightarrow V_x = V_{x1} + V_{x2}$

⊗ V_{x1} $\rightarrow 2 \angle -90^\circ$

Voltage division $V_{x1} = \frac{5}{10 + j5 + 5} \times 2 \angle -90^\circ$

$$= \frac{5}{10 + j5 + 5} \times 2 \angle -90^\circ = -0.2 - j0.6 \text{ V} = 0.63 \angle -108.4^\circ \text{ V}$$

⊗ V_{x2}

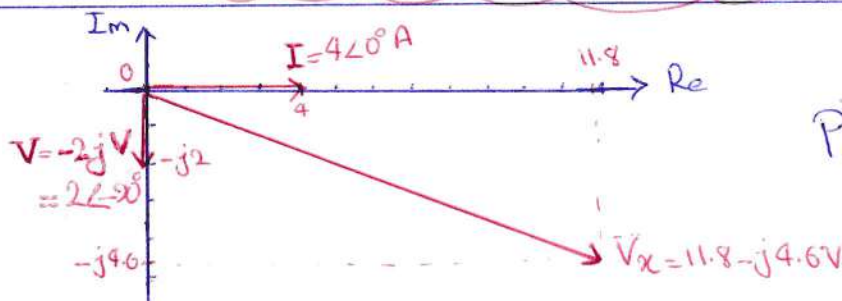
Source transformation is applicable also

Current division and Ohm's Law $V_{x2} = 5 I_x = 5 \times \frac{10}{10 + j5 + 5} \times 4 \angle 0^\circ$

$$= 12 - j4 \text{ V} = 12.65 \angle -18.93^\circ \text{ V}$$

$$\Rightarrow V_x = V_{x1} + V_{x2} = 11.8 - j4.6 = 12.66 \angle -21.29^\circ \text{ V}$$

(i) - b:



Phasor diagram

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Question 5 continued

(ii) [10 marks] Consider the following logical diagram shown in Figure 10.

- (4 marks) Derive and simplify the logical expression for Z.
- (4 marks) Construct the truth table relating Z to inputs A, B, and C.
- (2 marks) Using part a. and/or part b., show that Z can be realised using exactly one 3-input AND gate and one 1-input NOT gate showing all working.

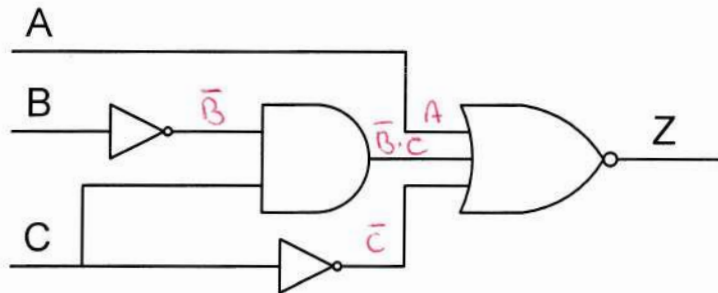


Figure 10

(ii)-a:

$$Z = A + \bar{B}.C + \bar{C}$$

DeMorgan's Theorem $\rightarrow Z = \overline{\bar{A} \cdot \bar{B}.C \cdot \bar{C}} = \bar{A} \cdot (\bar{B} + \bar{C}) \cdot C = \bar{A} \cdot (\bar{B}.C + \bar{C}.C) = \bar{A} \cdot (\bar{B}.C + 0) = \bar{A} \cdot \bar{B}.C$

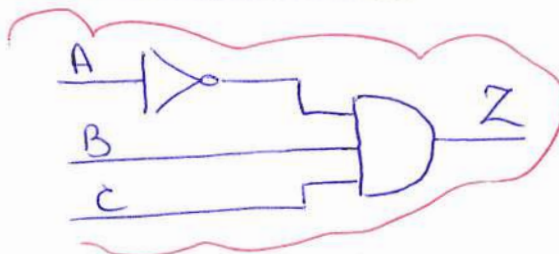
(ii)-b:

Truth Table:

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{B}.C$	$A + \bar{B}.C + \bar{C}$	$\overline{A + \bar{B}.C + \bar{C}}$	$\bar{A} \cdot \bar{B} \cdot C$
0	0	0	1	1	1	0	1	0	0
0	0	1	1	1	0	1	1	0	0
0	1	0	1	0	1	0	1	0	0
0	1	1	1	0	0	0	0	1	1
1	0	0	0	1	1	0	1	0	0
1	0	1	0	1	0	1	1	0	0
1	1	0	0	0	1	0	1	0	0
1	1	1	0	0	0	0	1	0	0

(ii)-c:

$$Z = \bar{A} \cdot \bar{B} \cdot C \Rightarrow$$



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